



Tracking-by-Clustering

A synthetic data set and an accuracy test of tracking-by-clustering

Multi-object tracking as one multicut problem on a spacetime graph, solved by greedy additive edge contraction

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Machine Learning for Computer Vision — Computer Vision I

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PART 1 · DANIEL MON

From Moving Spheres to an Instance

The task · motion model · data generation · gating · the multicut instance

The Task and Its Constraints

What the project brief fixes — Andres & Stein, “A synthetic data set and accuracy test of tracking-by-clustering”

Task — synthesize detections from moving spheres → build *one* tracking-by-clustering instance → solve → measure against sealed ground truth.

- ▶ **Geometry** — 3-torus spacetime $[0, L)^2 \times \mathbb{Z}_T$
- ▶ **No lifting** — edges only within and between *consecutive* frames
- ▶ **Deliverables** — VI + runtime, 2 motion models × noise levels

Data discipline — the tracker sees only positions + timestamps; labels stay sealed until evaluation.

Our scope — own greedy solver (no Gurobi); 2D space + cyclic time.

Motion Model

One stochastic engine, two regimes — inertial random walk and its ballistic limit

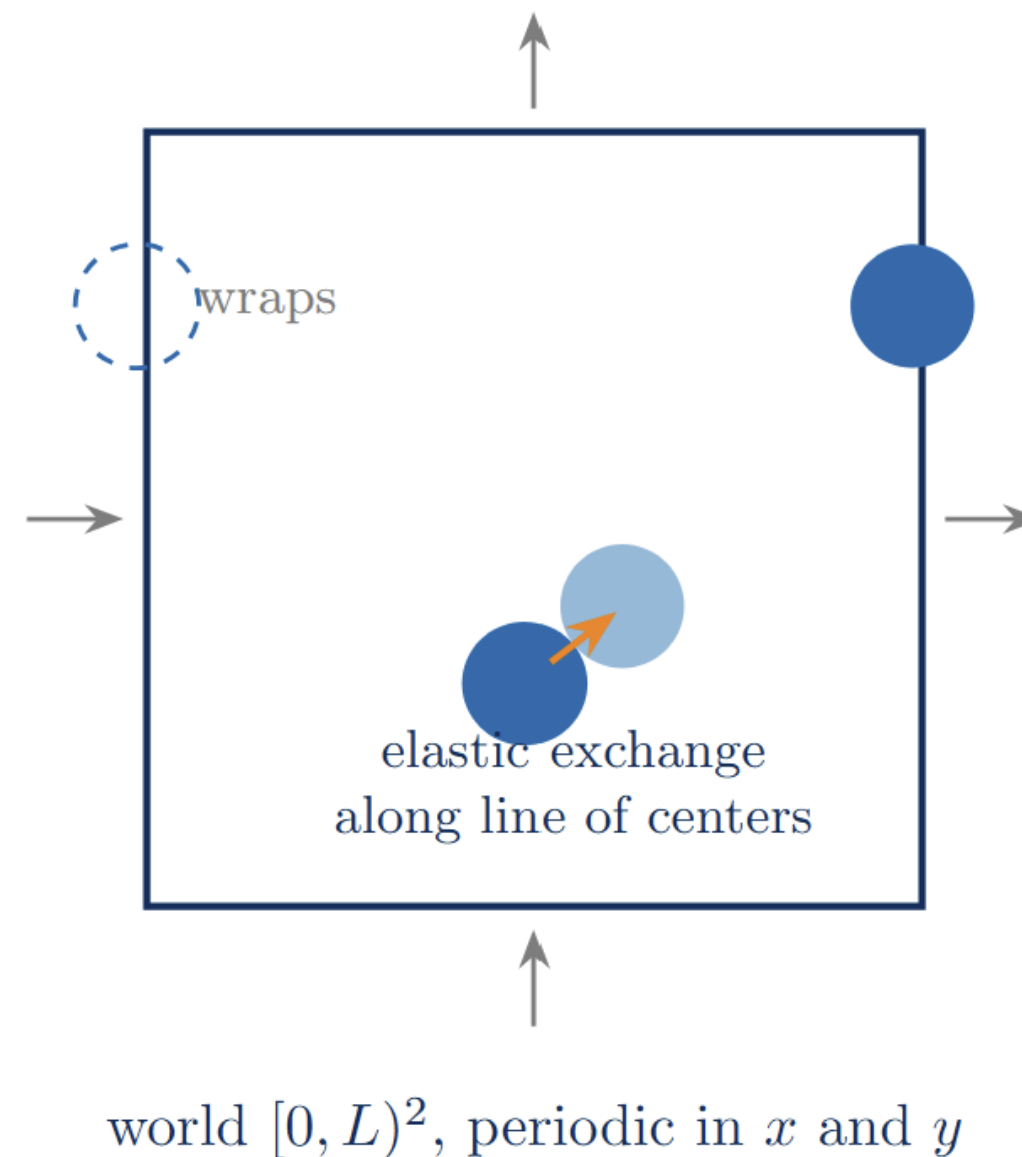
$$v(t+1) = v(t) + \eta(t), \quad \eta(t) \sim \mathcal{N}(0, \sigma_{\text{mot}}^2 I_2)$$
$$c(t+1) = \text{wrap}(c(t) + v(t+1) \Delta t, L)$$

Two regimes, one engine

- ▶ $\sigma_{\text{mot}} > 0$: **random walk** — direction drifts every frame
- ▶ $\sigma_{\text{mot}} = 0$: **ballistic** — straight lines between collisions

Elastic collisions

- ▶ trigger: torus distance $< 2r$, only if approaching $\langle v_i - v_j, \hat{n} \rangle > 0$
- ▶ equal-mass exchange along the line of centers



Torus rule

Every displacement is the minimum image:

$$\Delta_i = ((b_i - a_i + \frac{L}{2}) \bmod L) - \frac{L}{2}$$

Collisions work through the walls too.

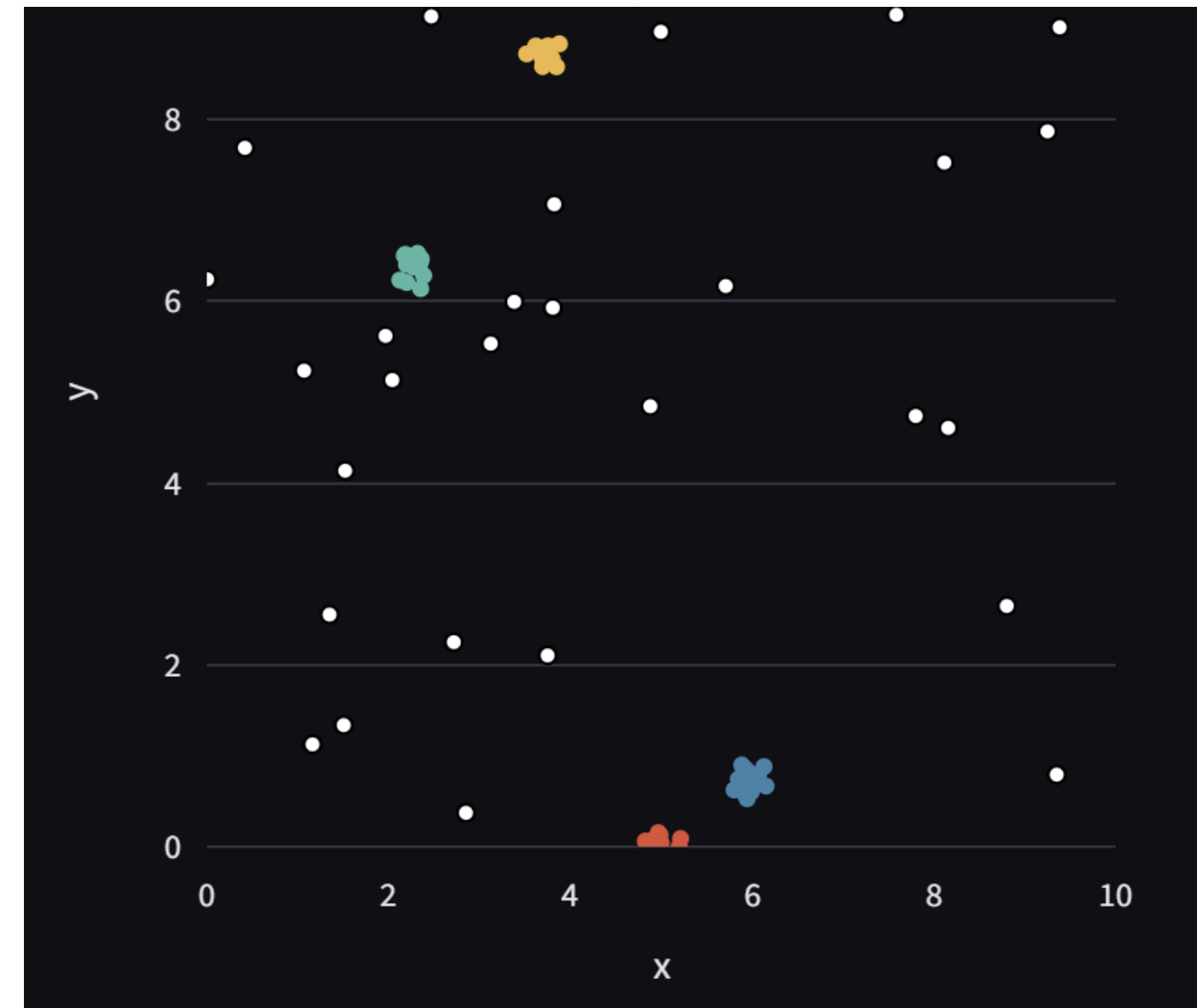
Implementation: `tbc/motion.py` + `tbc/collision.py` — one vectorized Euler step per frame.

Synthetic Data Generation

From hidden trajectories to the anonymous point cloud the tracker receives

1. **Input** — hidden trajectories $c(t, k)$ from the motion model.
2. **Sample inliers** — per sphere, per frame:
 n_{in} points $p = c(t, k) + \eta$,
 $\eta \sim \mathcal{N}(0, \sigma_{\text{obs}}^2 I)$, wrapped onto the torus.
3. **Add clutter** — n_{bg} points, uniform over the world, every frame.

Output: the anonymous cloud — positions + timestamps only; labels stay **sealed** until evaluation.



Instance Graph and Gating

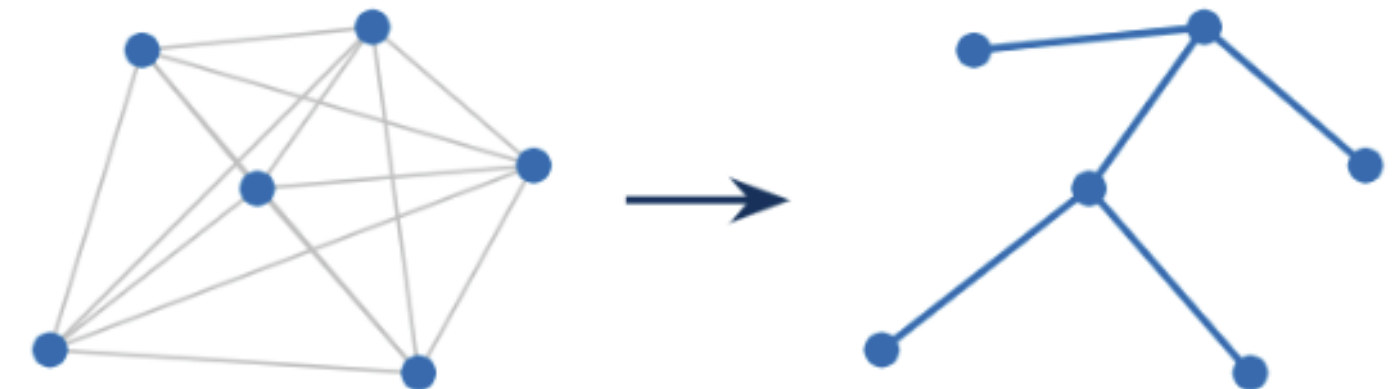
Complexity Problem- Gating

Naive graph: one edge per pair of points,

$$|E| = \binom{M}{2} \approx O(M^2)$$
$$M = 7800 \Rightarrow \approx 3 \times 10^7 \text{ edges}$$

Two distinct Problems:

- **Computational:**
Cost evaluation and GAEC ($O(|E| \log |E|)$)
on a quadratic edge set.
- **Statistical:**
Almost every pair is junk: each spurious
edge is a vote the solver must out-vote.



all pairs: $O(M^2)$

gated: $O(M)$

Gating: admit only edges between points that could *plausibly* belong to the same object — within a frame (ρ_{in}) and across consecutive frames (ρ_{mot}).

Result: $|E|$ grows **linearly** in M , and the graph arrives pre-cleaned.

Instance Graph and Gating

Gate Parameter Formula

- **Robust by construction** — clutter inflates mean and std; order statistics ignore any minority (50 % breakdown point).
- **Scale-equivariant** — rescale the cloud by λ , every ρ rescales itself. No simulation parameters: the motion gate needs no $v\Delta t$ term — displacement is *measured*, not modelled.

centre of the object mode
order statistic: clutter minority
cannot move it

width of the object mode
 $\text{med}_i |d_i - \text{med}|$
the robust std

$$\rho = \text{med}(\mathcal{D}) + c \cdot \text{MAD}(\mathcal{D})$$

ρ ← $\mathcal{D}$: **NN distances**
(torus metric)
within frame $\rightarrow \rho_{\text{in}}$
frames $t, t+1 \rightarrow \rho_{\text{mot}}$

c ← **margin, in widths**
 $c_{\text{gate}} = 3$ (admission)
 $c_{\text{cost}} = 2$ (decision boundary)

Instance Graph and Gating

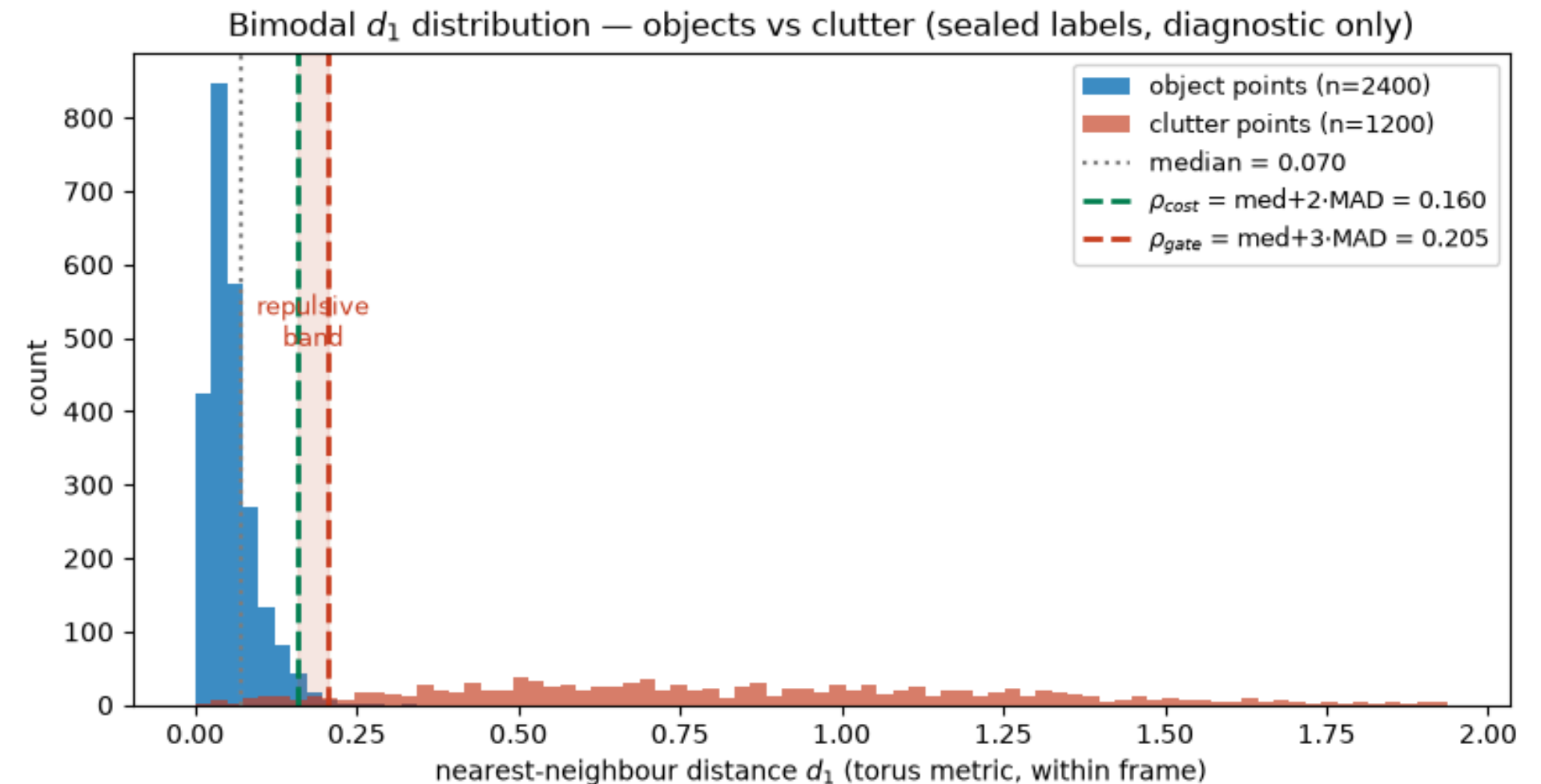
Gate Parameter Computation

1. Sample nearest-neighbour distances (torus metric):

- ▶ $\mathcal{D}_s$: each point $\rightarrow$ nearest neighbour *in its own frame*
- ▶ $\mathcal{D}_m$: each point in frame $t \rightarrow$ nearest neighbour *in frame $t+1$*

2. Estimate each radius robustly:

$$\rho = \text{med}(\mathcal{D}) + c \cdot \text{MAD}(\mathcal{D})$$



$K = 4 \text{ spheres} \times 15 \text{ inliers} = 60 \text{ object pts/frame} \cdot 30 \text{ background pts/frame} \cdot \sigma_{obs} = 0.1 \cdot T = 40 \text{ frames} \cdot M = 3600 \text{ points total}$

Instance Graph and Gating

Calibration of C parameter

Setup

6 conditions $\times$ 3 seeds

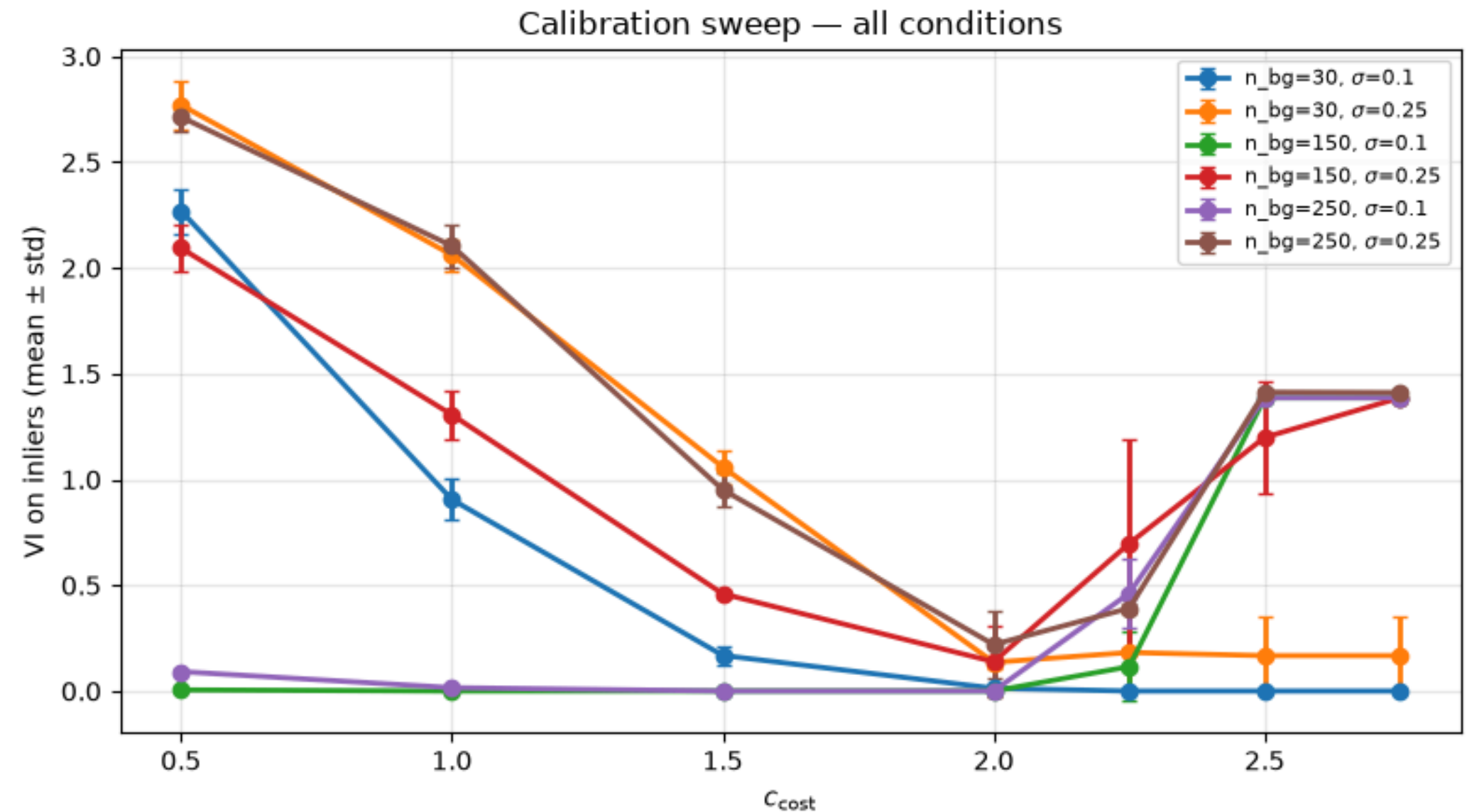
VI on inliers — pick the best worst case

The valley

left: *fragmentation*

right: *over-merging*

$c^* = 2.0$ — all conditions agree

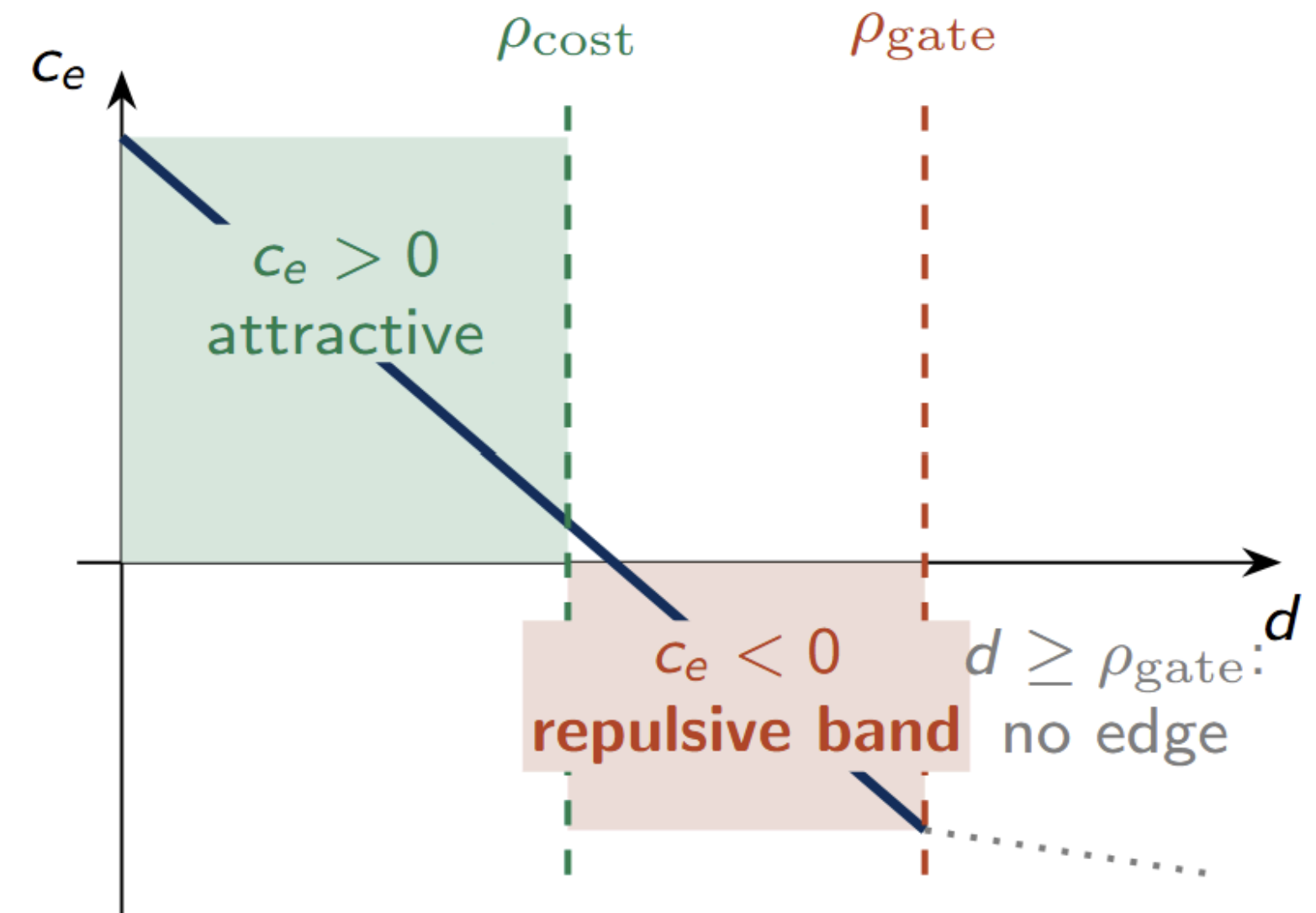


Cost Function

Decision Boundary Parameter

$$c_e = \alpha (\rho_{\text{cost}} - d_e), \quad \rho_{\text{cost}} < \rho_{\text{gate}}$$

- ▶ $\rho_{\text{gate}} = \text{med} + 3 \cdot \text{MAD}$
Admission (Recall): a dropped true edge is unrecoverable.
- ▶ $\rho_{\text{cost}} = \text{med} + 2 \cdot \text{MAD}$
Decision Boundary (Precision): a wrong vote can be out-voted.



PART 2 · MUHAMMAD TAHA

Solving, Measuring, Demonstrating

GAEC · implementation · constraint handling · evaluation & results · live demo

06

GAEC

the greedy idea

07

In practice

data structures, $O(E \log E)$

08

Constraints

held vs. relaxed — and why

09

Results

the noise sweep & its ceiling

10

Demo & wrap-up

Streamlit explorer, takeaways

Daniel handed us an instance — a gated spacetime graph with signed log-odds costs. My part: solve it, measure it, and show it live.

Greedy Additive Edge Contraction

Heuristic 3 — one greedy pass over the whole spacetime graph · [tbc/solver.py](#)

Key design decision: a track is one cluster of the entire graph — there is no separate “detect, then associate” stage.

1

Start: every point is its own cluster.

7,800 singleton clusters — nothing has been decided yet.

2

Maintain $\text{join}(a, b) = \Sigma$ of all edge costs between clusters a and b .

Within-frame and between-frame edges mix in one sum — the load-bearing idea.

3

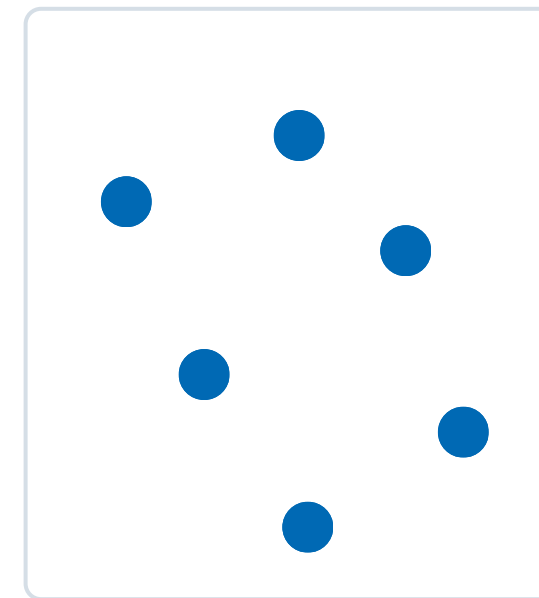
Contract the pair with the largest positive sum.

Fold the loser’s neighbor sums into the survivor, then repeat.

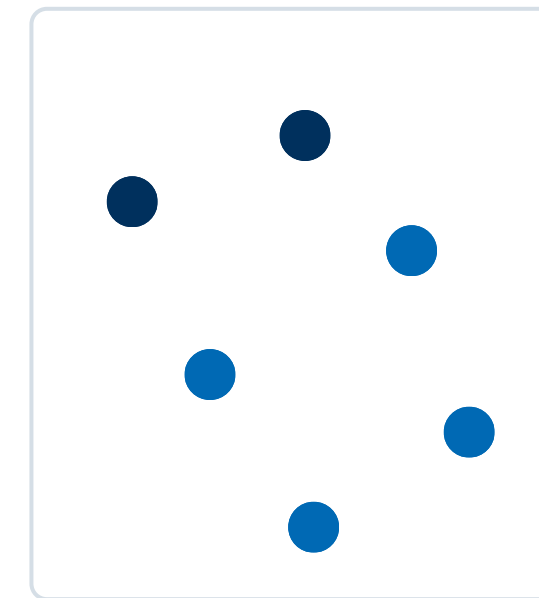
4

Stop when no positive pair remains.

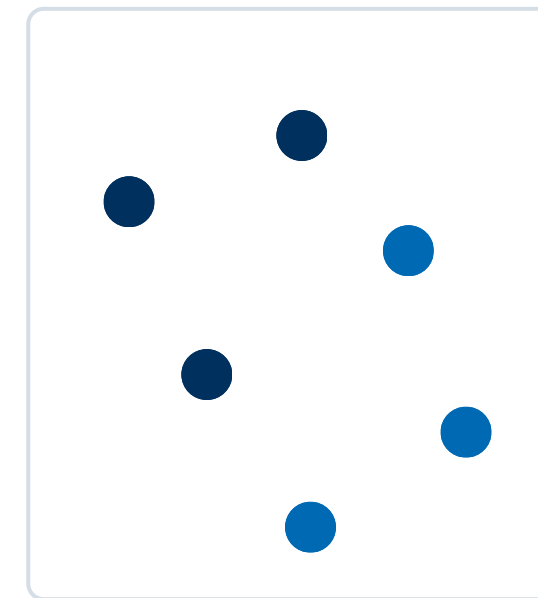
What is left is a partition — each cluster is one reported track.



every point its
own cluster



contract the best
positive pair



stop: no positive
pair left

dotted = candidate edge (positive cost) solid = contracted (inside one cluster)

Each contraction adds $\text{join}(a, b) > 0$ to the objective — the greedy is monotone and primal-feasible at every step.

GAEC in Practice

Data structures, complexity, and the structure of our instance · [tbc/solver.py](#)

1 Union–find with path compression

“Which cluster does this point belong to?” in near-constant amortized time — `find()` flattens the path as it walks it.

2 Lazy-deletion max-heap

Candidate pairs keep old priorities in the heap; stale entries are not removed eagerly — they are skipped when popped.

3 Per-cluster neighbor sums `join(a, b)`

Each cluster stores its running edge-cost sums. A contraction folds the loser’s map into the survivor’s in one pass.

Complexity: $O(|E| \log |E|)$ — full pipeline runs in ≤ 1.2 s (7,800 nodes).

Structure of our instance — a true multicut

Two radii per edge type, estimated from the data:

$\rho_{\text{gate}} = \text{med} + 3 \cdot \text{MAD}$ (admits candidate edges)

$\rho_{\text{cost}} = \text{med} + 2 \cdot \text{MAD}$ (zero-crossing of the cost)

$$c_e = \alpha (\rho_{\text{cost}} - d_e)$$

Edges with $\rho_{\text{cost}} < d_e < \rho_{\text{gate}}$ enter with $c_e < 0$:

$\approx 23\text{--}26\%$ of all edges are repulsive.

Contract while $\max \text{join}(a, b) > 0$; repulsive sums keep clusters apart at the boundary.

GAEC $\neq$ plain connected components — the baseline partition differs at all four noise levels; the repulsive edges decide the boundaries.

Constraint Handling

Heuristic4—which constraints hold by construction, which we relax, and why relaxing is safe

The exact IP demands three constraint families — our greedy treats each differently:

Transitivity — satisfied by construction



GAEC only ever merges, so the output is always a valid partition. No cycle inequalities are ever needed.

No-split / no-join — relaxed



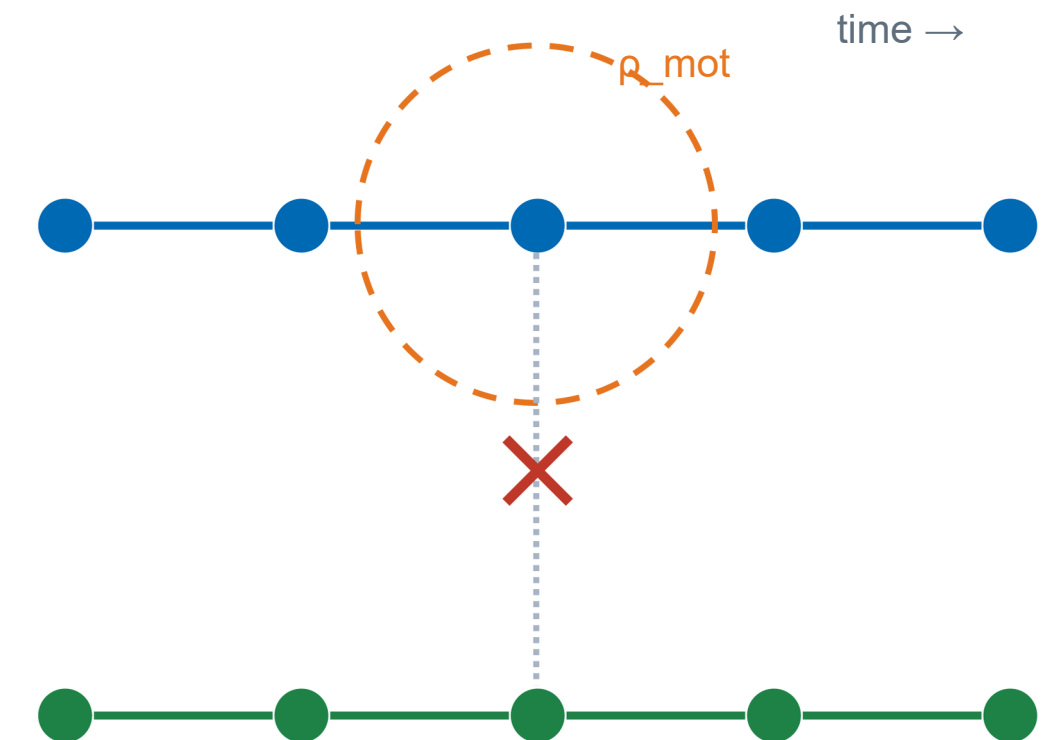
We solve the plain multicut version — the same trade-off Tang et al. (2016) make, so an efficient primal-feasible greedy applies. Stated explicitly in the report, never violated silently.

Why relaxing is safe here



A branching track would have to merge through the gate across object boundaries. At moderate noise the gates keep cross-object pairs disconnected — branching is unlikely in practice.

no-join, enforced by geometry



a join across tracks must first pass the gate — at moderate noise, it does not.

Evaluation Protocol

Variation of information on inliers, over a full motion × noise × seed grid

$$VI(T, P) = H(T \mid P) + H(P \mid T).$$

Note: T is true partition, P is predicated partition

$H(T \mid P)$ — merge error

when one predicted cluster blends two real tracks together — so looking at our cluster, you can't tell which true object it belonged to.

$H(P \mid T)$ — split error

When one real track gets shattered across several clusters — so knowing the true object, you still can't point to a single cluster that holds it.

A true metric — $VI = 0 \Leftrightarrow$ identical partitions; lower is better.

Headline number on inliers only — clutter has no track identity to recover; ARI and NMI reported as secondary metrics.

Experiment grid

$\sigma \in \{0.05, 0.10, 0.20, 0.40\} \times \{\text{ballistic, random walk}\} \times 3 \text{ seeds}$
 $\Rightarrow$ 24 full pipeline runs.

Per cell: VI mean $\pm$ std, ARI, NMI, split/merge decomposition, runtime.

What the sweep shows

At $\sigma \leq 0.10$ merge error is exactly zero — all residual error is split error from filtered singletons.

At $\sigma = 0.40$ both terms contribute $\approx$ equally; both motion models behave near-identically.

Results — The Noise Sweep

Near-exact recovery through $\sigma = 0.20$, then a mixed breakdown · 2 models × 4 noise levels × 3 seeds, inliers only

VI $\approx$ 0.09

at $\sigma = 0.05$ — all $K = 4$ tracks recovered (ARI ≈ 0.99); residual is a handful of filtered singletons

0.20 $\rightarrow$ 0.40

the breakdown window — VI rises from ≈ 0.4 – 0.5 to ≈ 1.5 ; ARI drops from ≥ 0.84 to ≈ 0.30

VI $\approx 1.5 > \ln 4$

split entropy adds on top of the fusion ceiling
— the breakdown is mixed, not a collapse to one cluster
7,800 nodes · ≈ 1.1 – 1.25×10^5 candidate edges at low σ , decreasing with σ (adaptive gates + core-point filter) · seeds {0, 1, 2} · split = $H(P \mid T)$, merge = $H(T \mid P)$.

model	σ	VI mean $\pm$ std	ARI	split	merge	t [s]
ballistic	0.05	0.091 $\pm$ 0.017	0.985	0.091	0.000	1.1
ballistic	0.10	0.185 $\pm$ 0.029	0.969	0.185	0.000	0.9
ballistic	0.20	0.509 $\pm$ 0.170	0.844	0.398	0.111	0.7
ballistic	0.40	1.528 $\pm$ 0.051	0.298	0.781	0.747	0.6
random walk	0.05	0.086 $\pm$ 0.015	0.986	0.086	0.000	1.1
random walk	0.10	0.175 $\pm$ 0.027	0.971	0.175	0.000	0.9
random walk	0.20	0.382 $\pm$ 0.021	0.934	0.379	0.003	0.8
random walk	0.40	1.507 $\pm$ 0.029	0.301	0.760	0.746	0.6

Results — Failure Anatomy

How the error decomposes as noise grows — and why the gates hold as long as they do

1 Low noise ($\sigma \leq 0.10$): zero merge error.

No tracks are ever fused. The residual VI is pure split error: the core-point filter (`min_neighbors = 3`) isolates a few dozen inlier points as singletons.

2 $\sigma = 0.20$: split error grows, structure holds.

Exactly $K = 4$ large clusters on every seed; $\approx 3,000$ clutter points end as singletons — repulsive edges and the filter leave no clutter bridges between tracks.

3 $\sigma = 0.40$: mixed breakdown, split $\approx$ merge ≈ 0.75 .

Diffuse clouds part-fragment (302 inlier singletons) and part-fuse — two dominant clusters of 3,536 and 1,154 points. VI ≈ 1.5 exceeds $\ln 4$.

4 The gates are calibrated on the data itself.

Nearest-neighbor distances mix the object scale with the clutter scale; the robust estimator reads the boundary between them from that mixture.

$\rho = \text{med}(D) + c \cdot \text{MAD}(D)$, $\text{MAD}(x) = \text{med } |x - \text{med}(x)|$ — robust to clutter: the median ignores the noise tail that inflates a k -std threshold.

Spin up a world — and watch it get tracked.

Every parameter is live: world size, spheres, motion model, noise, gate & cost radii (c_{gate} , c_{cost}), noise filter — the instance is rebuilt and re-solved as the sliders move.

- 1 **World animation** — the hidden ground truth — four spheres, elastic collisions, wrapping walls.
- 2 **Radar vs. truth** — what the tracker actually receives — anonymous points, clutter included.
- 3 **Instance graph** — the gates at work — attractive and repulsive edges inside ρ_{gate} .
- 4 **GAEC tracks** — the solved partition, colored by cluster — side by side with the truth.
- 5 **Noise sweep** — drag σ from 0.2 to 0.4 live and watch the tracks break apart — fraying first, then fusion.

```
$ streamlit run app.py → localhost:8501
```



app.py — the full pipeline behind a sidebar; the same tbc modules, no separate demo code.

Streamlit + Plotly — thin UI over the tbc package; results cached, re-solves instant.

Conclusions

What we built, what we measured, what we learned

- 1 One clustering problem tracks everything.** Synthesis $\rightarrow$ instance $\rightarrow$ GAEC $\rightarrow$ VI on the 3-torus — the tracker sees only positions p and timestamps t .
- 2 Two radii make it a true multicut.** The gate (med + 3·MAD) sets recall; the cost zero-crossing (med + 2·MAD) sets precision — $\approx \frac{1}{4}$ of all edges are repulsive, so the full greedy is load-bearing.
- 3 Near-exact recovery through $\sigma = 0.20$.** ARI ≥ 0.84 with exactly $K = 4$ large clusters on every seed and both motion models; merge error is zero at low noise.
- 4 The $\sigma = 0.40$ breakdown is mixed — and quantified.** Split and merge error contribute equally; VI ≈ 1.5 exceeds $\ln K$ because fragmentation adds entropy beyond the fusion ceiling.

Outlook learned edge costs · explicit no-split / no-join handling · comparison against the exact ILP · gate estimation in clutter-free regimes

Thank you

Questions are welcome